

Received 31 March 2026, accepted 19 April 2026, date of publication 23 April 2026, date of current version 30 April 2026.

Digital Object Identifier 10.1109/ACCESS.2026.3687059

RESEARCH ARTICLE

DP-FedHybrid: A Differentially Private Federated Stacking Framework Integrating Tree-Based and Transformer Models for Secure Heart Disease Prediction

ATTA UR RAHMAN¹, BIBI SAQIA², MUHAMMAD TAYYAB^{1,3}, MEHRAN ULLAH⁴,
MUHAMMAD SHAHID ANWAR¹, (Senior Member, IEEE), AND PIR NOMAN AHMAD¹

¹IRC for Finance and Digital Economy, KFUPM Business School, King Fahd University of Petroleum and Minerals (KFUPM), Dhahran 31261, Saudi Arabia

²Department of Computer Science, University of Science and Technology Bannu, Bannu, Khyber Pakhtunkhwa 28100, Pakistan

³Information Systems and Operations Management Department, KFUPM Business School, King Fahd University of Petroleum and Minerals, Dhahran 31261, Saudi Arabia

⁴School of Business and Creative Industries, University of the West of Scotland, PA1 2BE Paisley, U.K.

Corresponding authors: Atta Ur Rahman (attaur.rahman@kfupm.edu.sa) and Mehran Ullah (mehran.ullah@uws.ac.uk)

ABSTRACT Heart disease is rapidly increasing. For early and accurate prediction, advanced machine learning (ML) models are needed. However, the sensitive nature of clinical data, along with strict regulatory constraints, creates significant challenges to traditional centralized learning systems. Federated learning (FL) has emerged as a decentralized paradigm for collaborative model training, enabling multiple clients to jointly learn a global model without exchanging raw data. However, various existing FL frameworks rely on single-model architectures. They struggle to capture complex feature interactions in tabular healthcare data. To address these limitations, we propose a novel DP-FedHybrid, differentially private federated stacking framework that integrates heterogeneous models within a secure and decentralized learning architecture. We use a multi-client FL setup, where each client independently trains CatBoost and Transformer models in parallel, and their predictive outputs are combined through a stacking-based meta-learning mechanism. The Transformer component is optimized using differentially private stochastic gradient descent (DP-SGD), incorporating gradient clipping and calibrated Gaussian noise injection to ensure formal privacy guarantees. To further strengthen, we use a stacking-based meta-learning layer that aggregates probabilistic outputs from client-side models. It enables effective knowledge fusion and enhances robustness and generalization under non-independent and identically distributed (non-IID) data. The proposed framework is evaluated on a benchmark heart disease dataset, where we obtained an accuracy of 95.12% under standard FL and 94.15% under DP constraints, outperforming closely related works. The proposed work advances the existing literature by providing a scalable, hybrid, and privacy-preserving FL paradigm for heart disease prediction.

INDEX TERMS DP-FedHybrid, DP-SGD, hybrid models, meta-learning, non-IID.

I. INTRODUCTION

Heart disease is rapidly increasing worldwide, making accurate and early prediction crucial for effective clinical analysis [1]. However, heart disease data is complex [2]. The increasing digitization of medical records and the proliferation

of multi-institutional healthcare data have created unique opportunities for predictive modeling [3]. However, these advancements are accompanied by significant challenges, primarily due to the sensitive nature of clinical data [4]. In the Internet of Medical Things (IoMT), doctors treat a patient's illness by using artificial intelligence (AI) systems to analyze medical data gathered via mobile devices. However, due to design limitations, centralized AI systems fail to provide

The associate editor coordinating the review of this manuscript and approving it for publication was Chao Tong¹.

distributed learning and maintain patient data safely [5]. ML has been widely used to find cardiovascular disease (CVD) risk based on various features [6]. However, the privacy issues make it difficult to perform collaborative training to predict CVD using ML models [7]. In practice, healthcare datasets are inherently decentralized, residing across multiple hospitals and institutions [8]. The dataset is often heterogeneous, high-dimensional, and exhibits non-IID characteristics [9]. These characteristics make centralized ML approaches problematic and misaligned with privacy requirements. Therefore, there is an increasing demand for collaborative, data-driven intelligence in healthcare [10]. However, the requirements to preserve patient privacy remain an obstacle for the classic ML approach [11]. To address these challenges, existing literature has adopted FL as a decentralized paradigm that enables collaborative model training without sharing raw data [12]. FL offers comprehensive multi-center processing by enabling models to be trained in a distributed fashion that protects patients' privacy [13]. In FL, a global model is distributed to multiple data-holding sites, where local training is performed on each site's private data. The locally updated models are then sent back to a central server, where they are aggregated to update the global model. The trained model weights from each participant site are averaged on this server before a new training cycle begins [14].

Previous studies have demonstrated that FL can effectively mitigate privacy risks while maintaining competitive performance compared to centralized approaches [15]. However, existing FL frameworks predominantly rely on homogeneous model architectures, which limit their ability to capture diverse feature interactions in tabular healthcare data [16]. Also, the deep federated models often suffer from instability, overfitting, and sensitivity to limited data. To overcome these challenges, we propose DP-FedHybrid, a novel differentially private federated stacking framework that integrates tree-based and transformer-based learning within a unified and privacy-preserving architecture. The proposed approach employs a multi-client FL setup to simulate realistic healthcare data distributions, where each client locally trains a hybrid CatBoost and Transformer model. Where the CatBoost is utilized to capture complex non-linear feature interactions and generate enriched tabular representations, which are subsequently leveraged by the Transformer component to model higher-order dependencies. The Transformer is optimized using DP-SGD, incorporating gradient clipping and calibrated Gaussian noise injection to ensure formal privacy guarantees while maintaining training stability. Model updates from the Transformer component are aggregated across clients using the Federated Averaging (FedAvg) algorithm to construct a global representation under decentralized constraints. To further enhance predictive performance, a stacking-based meta-learning layer is used, which aggregates probabilistic outputs from client-side models and learns an optimal combination through a logistic regression (LR) meta-learner. The framework is evaluated on

a heart disease dataset using both standard FL, and DP-FL to assess predictive performance, robustness, and privacy-utility trade-offs. The experimental results demonstrate that the integration of hybrid modeling significantly enhances representational capacity, while the stacking-based meta-learning layer improves generalization under non-IID data distributions. The following are the key contributions of the proposed work.

- **Hybrid Federated Architecture:** We develop a novel unified framework that integrates tree-based and transformer-based models within an FL setup, overcoming the limitations of homogeneous model designs for tabular heart disease data.
- **Privacy-Preserving Optimization:** We incorporate DP-SGD with gradient clipping and noise injection into transformer-based federated training, achieving strong privacy guarantees with minimal performance degradation.
- **Stacking-Based Meta-Learning:** We use a meta-learning ensemble layer that aggregates client-level predictions, enhancing generalization, robustness, and cross-client knowledge fusion under non-IID data distributions.
- **Scalable and Robust Framework:** We develop a scalable, high-performance, and privacy-aware learning paradigm tailored for distributed heart disease prediction, advancing the state-of-the-art in secure FL environments.

The main objective of this work is to design and verify a scalable system that makes it easier for complex ML models to be trained among distributed clients, eliminating the security risks related to the centralized collection of private patient data. The remaining sections of this study are organized as follows. Section II reviews the literature on heart disease prediction using centralized learning, FL, DP, and their combinations. This section also explains the current literature and identifies the research gaps in FL-based heart disease prediction. Section III explains the proposed DP-FedHybrid framework, detailing the hybrid CatBoost, Transformer architecture, DP integration, federated optimization, and stacking-based meta-learning. Section IV explains the experiments conducted using the federated setup and implementation details. Section V explains the model's performance for various client configurations and ablation studies, using standard evaluation metrics. This section covers the quantitative performance and comparative analysis. Section VI explains the key findings, emphasizing robustness, privacy-utility trade-offs, and model behavior. Finally, Section VII concludes this study with potential applications and directions for future research.

II. RELATED WORK

Early detection of heart disease is crucial in the medical domain, as it enhances treatment and improves patient survival rates [17]. Various studies have been conducted on heart disease prediction to enhance clinical outcomes.

The study in [18] developed an AI framework that extracts electrocardiogram (ECGs) features and uses wavelet transformations for heart disease prediction. They obtained better performance for disease prediction and assisted cardiologists in various decision making process. They highlighted the effectiveness of automated feature learning in clinical diagnosis. AI methods have been used in various cases for heart disease prediction to provide speedy and effective pre-screening [19]. In order to create an accurate ML system for early cardiac disease prediction, the study in [20] employed various feature selection techniques. They applied techniques like mutual information, ANOVA, and chi-square. They used classification techniques such as support vector machines (SVM), XGBoost, bagging, decision trees (DT), and random forests (RF). The study detected heart disease at an early stage using ML approaches [21]. They compared nine ML algorithms and suggested a new Principal Component Heart Failure (PCHF) feature engineering approach to choose the most significant features and improve performance. The study in [22] used a stacked ensemble model combining multiple ML algorithms optimized through advanced search techniques for accurate heart disease prediction. The study used a TabNet model in combination with catBoost to increase the precision and effectiveness of heart disease prediction [23]. Their technique enables instantaneous processing, allowing for quick predictions based on streaming of IoMT data. Traditional ML approaches for heart disease prediction have largely relied on centralized training paradigms [24]. These approaches have achieved promising predictive performance. However, they assumed centralized data availability, which is often limited in real-world healthcare settings due to privacy constraints. Also, collecting large volumes of healthcare data in a single location is challenging, because medical data often exists in different hospitals, labs, and healthcare centers [25].

To address data decentralization and privacy concerns, FL has been widely adopted as a distributed training paradigm [26]. By training on a wider variety of data that is distributed across several devices or locations, FL models are able to capture a wider range of features, increasing their generalizability for real-world applications [27]. The problem of privacy-preserving predictive modeling in healthcare has attracted significant attention in recent years. Early FL frameworks, such as FedAvg, demonstrated that collaborative model training across multiple clients can be achieved without sharing raw data, thereby preserving data locality. The work done in [28] developed an FL-based DL model for heart disease detection. After incorporating a DP method and adding random noise to avoid information leakage, the generated local updates are sent to the global model. For the categorization of arrhythmias in ECGs, a weighted FL method is developed [29]. Data variability across clients is considered, with each client's weight dynamically adjusted based on its contribution to overall model improvement. The study conducted in [30] used the FL framework to address data privacy concerns by incorporating DP, resulting in secure and effective model aggregation. The study in [31] used

the Federated Explainable AI (FedXAI) framework, which collected health data on consumer devices while only model updates were shared for collaborative learning. Explainability ensures that medical professionals and patients both have an understanding of how the model makes predictions. Using decentralized healthcare data, the study in [32] used an FL architecture with graph neural networks (GNNs) to provide safe and effective heart attack risk prediction. To preserve data locality and protect patient privacy, they divided the preprocessed data across three clients to create graph-structured representations.

Previous studies have used FL for various heart disease prediction tasks. They have shown that FL can maintain performance comparable to centralized models while improving data privacy [33]. However, most existing FL approaches primarily utilize homogeneous model architectures, particularly deep neural networks, which limits their effectiveness for structured tabular data. Moreover, FL systems often face challenges related to statistical heterogeneity (non-IID data), client drift, and communication inefficiencies, which degrade model convergence and generalization performance [34].

The proposed DP-FedHybrid framework addresses these limitations by introducing a novel hybrid and privacy-preserving FL architecture designed for tabular heart disease data. This hybrid design enhances representational capacity and mitigates the weaknesses of individual models. To further strengthen the model, DP is incorporated specifically within the transformer optimization process using DP-SGD with gradient clipping and calibrated Gaussian noise, enabling formal privacy guarantees while maintaining model stability and performance. Instead of relying solely on parameter averaging, the proposed framework aggregates probabilistic outputs from client-side models and learns an optimal combination through meta-learning. This approach improves robustness to non-IID data distributions and enhances generalization by effectively fusing knowledge across heterogeneous clients.

III. MATERIALS AND METHODS

In this work, we use a novel DP-FedHybrid framework, which is a multi-client FL setup to simulate decentralized heart disease data distributions, as shown in Figure 1. Each client locally trains a hybrid model integrating CatBoost and a Transformer network, where CatBoost captures non-linear feature interactions and generates enriched representations that are further processed by the Transformer to model higher-order dependencies. The Transformer component is optimized using DP-SGD, incorporating gradient clipping and calibrated Gaussian noise to ensure privacy. During each communication round, locally trained Transformer parameters are aggregated using the FedAvg algorithm to construct a global model. We use probabilistic outputs from both CatBoost and Transformer models across clients, which are collected to form meta-features. A stacking-based meta-learning layer, implemented via LG, is then trained on these outputs to produce final predictions. This architecture

enhances representational capacity, privacy preservation, and generalization under non-IID conditions.

TABLE 1. Stratified multi-client data distribution.

Client	Sample	Positive	Negative	Positive Ratio
Client 1	342	167	175	0.48
Client 2	342	184	158	0.53
Client 3	341	175	166	0.51

A. DATASET AND PREPROCESSING

We used the UCI heart disease dataset [35], which is widely used as a benchmark dataset for cardiovascular risk prediction. The dataset consists of clinical records collected from multiple institutions. It contains 1025 instances with 14 key attributes, such as age, gender, chest pain type, blood pressure (BP), cholesterol, and electrocardiographic (ECG) measurements, which are used to predict the presence of heart disease as a classification task. The dataset is heterogeneous and includes both categorical and continuous features. In this work, the dataset is first preprocessed and partitioned into a multi-client federated setting, simulating decentralized hospital environments. The dataset is partitioned using a stratified multi-client strategy, ensuring approximately equal sample sizes and consistent class distributions across clients, as shown in Table 1. This setup maintains statistical representativeness while introducing heterogeneity, thereby providing a realistic and well-balanced non-IID environment for FL evaluation. Each client locally trains the proposed hybrid CatBoost and Transformer model, where CatBoost captures non-linear feature interactions and the Transformer models higher-order dependencies. The target variable is binarized for classification, and the data is further split into training and testing sets.

B. MODEL DEVELOPMENT

In this work, we develop a novel framework that integrates CatBoost and Transformer within a privacy-preserving FL environment. The data is first partitioned across multiple clients, where each client independently trains both models on local data. The tabular heart disease dataset is described in Eq. (1), where $x_i \in \mathbb{R}^d$ represents the feature vector and $y_i \in \{0, 1\}$ denotes the binary class label.

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N \quad (1)$$

1) FL SETUP

Our proposed framework simulates an FL environment by performing a split of the training data across $K = 3$ decentralized clients as described in Eq. (2). Each client receives a distinct subset of the data while preserving the original class distribution, described in Eq. (3), ensuring balanced learning across clients. Within each client, a hybrid local model is trained, consisting of two complementary components: a CatBoost classifier and a Transformer-based neural network. The CatBoost model is described in Eq. (4),

where $h_m(x)$ denotes the output of m -th decision tree and η is the learning rate. The CatBoost is configured with a fixed number of boosting iterations $M = 300$ and depth to efficiently capture structured, tabular feature interactions. The Transformer component is trained for a limited number of epochs to model complex, non-linear relationships and contextual dependencies within the data. The Transformer model at each client is a lightweight residual feed-forward tabular network. For the input feature vector $x \in \mathbb{R}^d$, the first operation is a linear embedding described in Eq. (5), where $W_e \in \mathbb{R}^{p \times d}$, $b_e \in \mathbb{R}^p$.

$$\mathcal{D} = \bigcup_{k=1}^K \mathcal{D}_k, \quad \mathcal{D}_k \cap \mathcal{D}_j = \emptyset, \quad k \neq j \quad (2)$$

$$\mathcal{D}_k = \{(x_i^{(k)}, y_i^{(k)})\}_{i=1}^{n_k} \quad (3)$$

$$f_k^{CB}(x) = \sigma \left(\sum_{m=1}^M \eta h_m(x) \right) \quad (4)$$

$$z^{(0)} = W_e x + b_e \quad (5)$$

For each residual block $\ell = 1, \dots, L$, where $L = \text{CFG.depth} = 2$, the hidden representation is updated as described in Eq. (6), where $\phi_\ell(\cdot)$ is a feed-forward block, described in Eq. (7).

$$z^{(\ell)} = z^{(\ell-1)} + \phi_\ell(z^{(\ell-1)}) \quad (6)$$

$$\phi_\ell(z) = \text{Dropout}(\text{ReLU}(W_\ell z + b_\ell)) \quad (7)$$

The final prediction is obtained via a sigmoid activation applied to the output logit as described in Eq. (8), and the probability output is described in Eq. (9), where θ_k denotes all trainable parameters.

$$a_k(x) = w_h^\top z^{(L)} + b_h \quad (8)$$

$$f_k^{\text{TF}}(x; \theta_k) = \sigma(a_k(x)) = \frac{1}{1 + e^{-a_k(x)}} \quad (9)$$

The Transformer is optimized locally using a binary cross-entropy loss function over mini-batches. For a mini-batch $\mathcal{B}_k = \{(x_i, y_i)\}_{i=1}^B$ sampled from client k , the Transformer is trained using binary cross-entropy with logits described in Eq. (10), where $a_i = a_k(x_i)$. The gradient of the local loss is described in Eq. (11).

$$\mathcal{L}_k(\theta_k; \mathcal{B}_k) = -\frac{1}{B} \sum_{i=1}^B [y_i \log \sigma(a_i) + (1 - y_i) \log (1 - \sigma(a_i))] \quad (10)$$

$$g_k = \nabla_{\theta_k} \mathcal{L}_k(\theta_k; \mathcal{B}_k) \quad (11)$$

2) DIFFERENTIAL PRIVACY

We incorporate a DP mechanism to mitigate potential information leakage from model updates. The Transformer component is trained using DP-SGD, where gradients are subjected to norm clipping and calibrated Gaussian noise injection at the client level before transmission. This process reduces the risk of reconstructing sensitive information from shared updates. In this work, the randomized mechanism $\mathcal{A}$ is

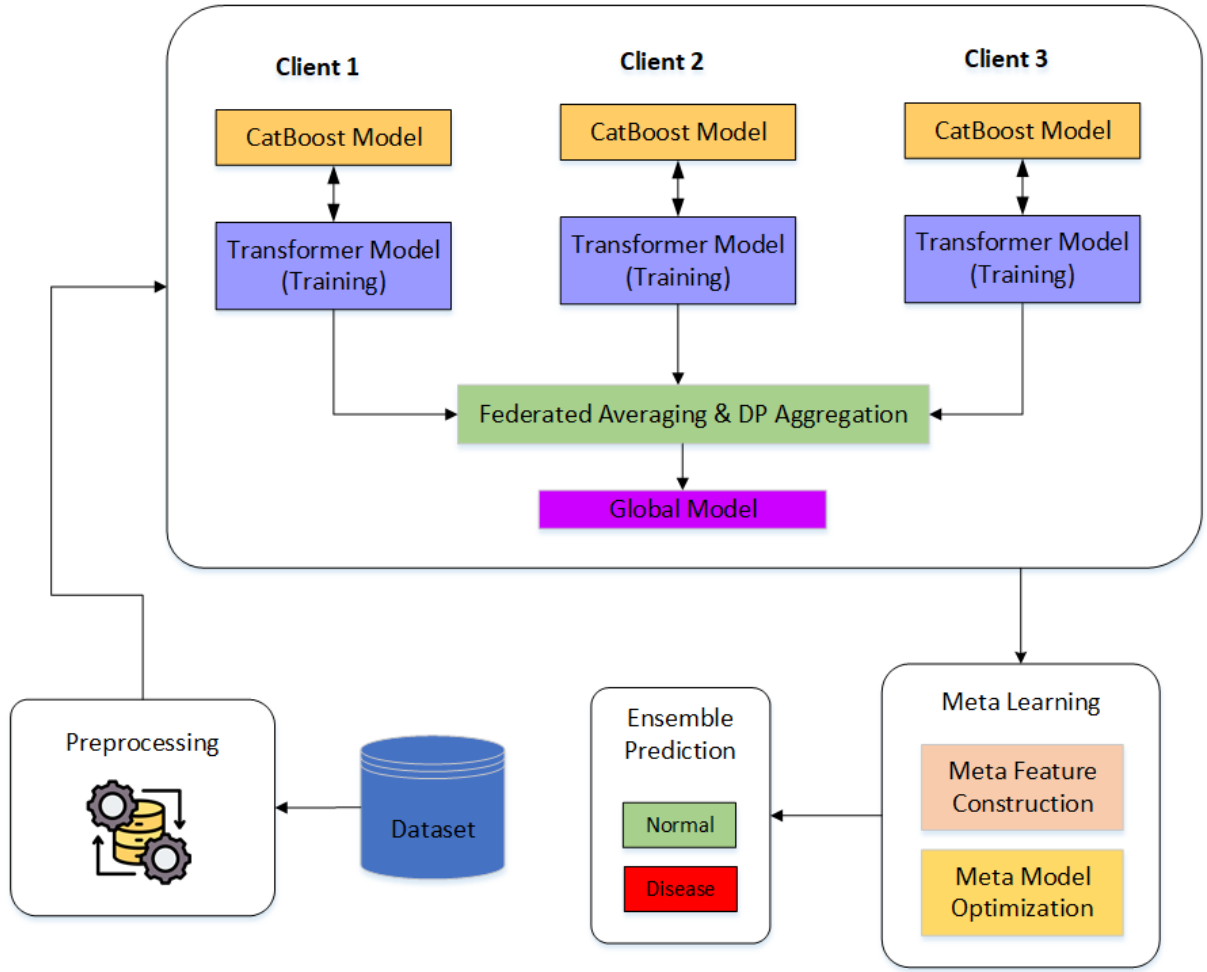


FIGURE 1. Overview of the proposed method, using FL, DP, and meta-Learning.

(ϵ, δ) differentially private if for any two neighboring datasets D and D' differing in one record, and for any measurable output set S , as described in Eq. (12).

$$\Pr[A(D) \in S] \leq e^\epsilon \Pr[A(D') \in S] + \delta \quad (12)$$

In this work, the privacy is approximated through gradient clipping and Gaussian perturbation during each local optimization step, as described in Eq. (13), which corresponds to Eq. (14), where $g_{k,r}$ denotes the gradient tensor for parameter block r , and R is the number of trainable tensors.

$$\|g_k\|_2 = \left(\sum_{r=1}^R \|g_{k,r}\|_2^2 \right)^{1/2} \quad (13)$$

$$\|g_k\|_2 = \left(\sum_{p \in \theta_k} \sum_j (p \cdot \text{grad}_j)^2 \right)^{1/2} \quad (14)$$

The gradient clipping mechanism is a crucial component in differentially private optimization, as it explicitly controls the sensitivity of the learning algorithm to individual data samples. To bound sensitivity, gradients are clipped to a

maximum norm C , where $C = \text{CFG.dp_clip} = 1.0$. During the experiments, it was observed that large gradient magnitudes negatively impact DP optimization, as they increase sensitivity, require aggressive clipping, and amplify the effect of noise injection, leading to degraded convergence and performance. The clipping coefficient is described in Eq. (15), where $\epsilon > 0$ is a small numerical constant. The clipped gradient becomes $\bar{g}_k = \alpha_k g_k$, which ensures, $\|\bar{g}_k\|_2 \leq C$. It ensures that the maximum possible change in the gradient (due to the addition or removal of a single data point) is bounded.

$$\alpha_k = \min \left(1, \frac{C}{\|g_k\|_2 + \epsilon} \right) \quad (15)$$

After clipping, Gaussian noise is added to each gradient tensor described in Eq. (16), where I is the identity covariance matrix, indicating that the noise is isotropic (independent and identically distributed across all dimensions of the gradient tensor). The σ is the noise multiplier, which controls the magnitude of the injected noise, here $\sigma = 0.02$.

$$\tilde{g}_{k,r} = \bar{g}_{k,r} + \mathcal{N}(0, \sigma^2 C^2 I) \quad (16)$$

The next step is to perturb this bounded gradient by adding a random noise term ξ_r , which is sampled from a multivariate Gaussian distribution, $\xi_r \sim \mathcal{N}(0, (\sigma C)^2 I)$. The perturbed (noisy) gradient is then computed as described in Eq. (17). This means that the true (clipped) gradient is intentionally corrupted with controlled randomness before being used for learning. The purpose of this is not to improve optimization directly but to ensure privacy protection.

$$\tilde{g}_{k,r} = \bar{g}_{k,r} + \xi_r \quad (17)$$

This perturbed gradient is then used for optimization. At the DP-STG step, the update is performed using the perturbed gradients, ensuring that privacy is enforced at every optimization step, as described in Eq. (18), where $\eta = \text{CFG.lr} = 10^{-3}$, and $\tilde{g}_k^{(t)}$ is the clipped and noise-perturbed gradient at iteration t . This equation shows that instead of using the true gradient, the model updates rely on a noisy approximation, thereby preventing direct leakage of information about individual data points.

$$\theta_k^{(t+1)} = \theta_k^{(t)} - \eta \tilde{g}_k^{(t)} \quad (18)$$

In this work, we use the Adam optimizer, which enhances standard SGD by incorporating adaptive learning rates through first and second-moment estimates of the gradients. In the DP setting, Adam operates on the noisy gradients, meaning privacy is preserved before any optimization logic is applied.

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) \tilde{g}_k^{(t)} \quad (19)$$

The first moment estimate (mean of gradients) is computed in Eq. (19), where β_1 controls the exponential decay rate of past gradients. This term acts like a momentum component, smoothing updates over time.

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) (\tilde{g}_k^{(t)})^2 \quad (20)$$

The second moment estimate (uncentered variance) is described in Eq. (20), where β_2 controls how past squared gradients are accumulated. This helps scale updates according to gradient variability.

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}, \quad \hat{v}_t = \frac{v_t}{1 - \beta_2^t} \quad (21)$$

Since both m_t and v_t are initialized at zero, they are biased toward zero in early iterations. To update this, bias-corrected estimates are computed in Eq. (21). After that, the parameter update rule is described in Eq. (22), where τ is a small stabilization constant.

$$\theta_k^{(t+1)} = \theta_k^{(t)} - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \tau} \quad (22)$$

The privacy is injected at the gradient level before the parameter update. This design ensures that every parameter update is both privacy-compliant and statistically meaningful, enabling effective training under strict privacy constraints.

In the local training phase within each federated communication round $r \in \{1, \dots, R\}$, where $R = \text{CFG.rounds} =$

5, each client initializes its local Transformer from the current global model. Then, for $E = \text{CFG.local_epochs} = 2$ local epochs, the client k optimizes its Transformer on D_k , producing updated parameters $\theta_k^{(r)}$. In parallel, the client trains a CatBoost model $f_k^{\text{CB},(r)}$, from its own local data. Thus, after round r , each client contributes a local CatBoost predictor and a locally updated, privacy-preserving Transformer parameter set.

Figure 2 illustrates the DP-FedHybrid framework, where heart disease data is stratified across three clients in a federated setup. Each client performs parallel CatBoost and Transformer training with DP-SGD for privacy preservation. Transformer parameters are aggregated via FedAvg to form a global model, while predictions from both models are combined through a stacking-based meta-learner. The proposed model iteratively refines performance across rounds and is evaluated using standard classification metrics.

3) FEDERATED AVERAGING

In this step, the knowledge learned independently by multiple clients is combined into a single global model without sharing raw data. This step is central to FL, as it enables collaborative training across decentralized datasets. Here, the $\theta_k^{(r)}$ denotes the final Transformer parameters from the client k at round r . The new global model is computed as the arithmetic mean described in Eq. (23), where each client contributes equally to the global model, regardless of the size of its dataset.

$$\theta^{(r)} = \frac{1}{K} \sum_{k=1}^K \theta_k^{(r)}. \quad (23)$$

After aggregation, the resulting parameter set $\theta^{(r)}$, describes the updated global Transformer model $f_{\text{TF}}(x; \theta^{(r)})$. This global model integrates knowledge from all clients and is then redistributed to them in the next communication round. This entire process preserves privacy because only model parameters are shared, and these parameters have already been trained using differentially private gradients. The global model benefits from collective learning while ensuring that no raw data or sensitive information is exposed. After aggregation, the server stores the updated global model and the set of client-specific CatBoost and Transformer models used for stacking, described in Eq. (24).

$$\mathcal{P}^{(r)} = \left\{ \left(f_1^{\text{CB}}, f_1^{\text{TF}} \right), \dots, \left(f_K^{\text{CB}}, f_K^{\text{TF}} \right) \right\} \quad (24)$$

4) META-LEARNING

In this step, we use the meta-feature space, which enables stacking-based learning in the proposed DP-FedHybrid framework. Instead of making predictions directly from a single model, the system first transforms model outputs into a new feature representation that captures multi-model and multi-client knowledge. The next stage constructs a meta-feature representation. For any input sample x , each client k generates two probability scores. CatBoost probability in Eq. (25), and Transformer probability in Eq. (26). These

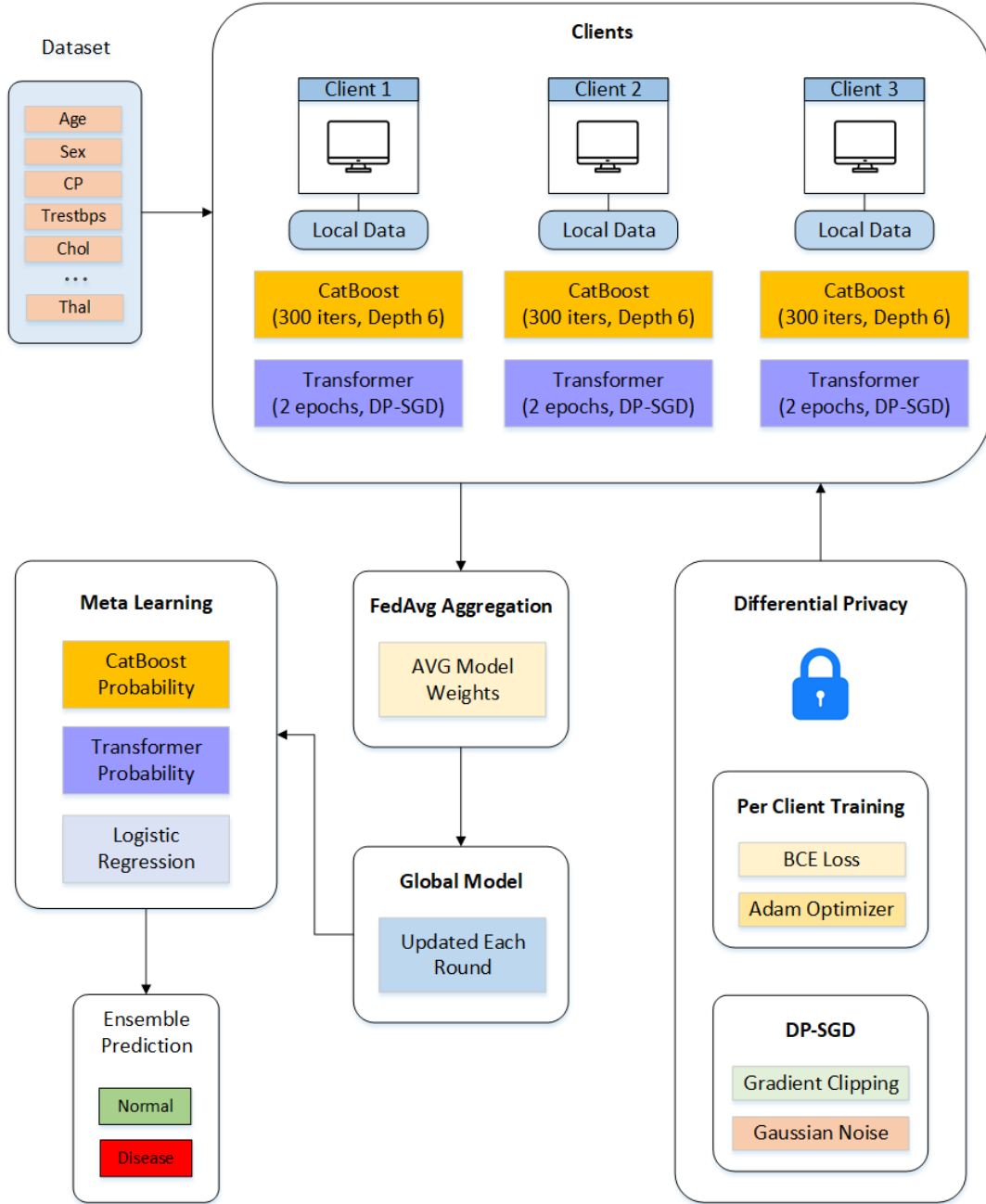


FIGURE 2. Proposed DP-FedHybrid model with FL, DP-SGD, and stacking-based meta-learning.

probabilities are model-specific and client-specific, meaning each reflects a different learned perspective on the same input.

$$p_k^{\text{CB}}(x) = f_k^{\text{CB}}(x) \quad (25)$$

$$p_k^{\text{TF}}(x) = f_k^{\text{TF}}(x; \theta_k) \quad (26)$$

These are concatenated into a meta-feature vector, described in Eq. (27). Since there are $K = 3$ clients and two predictors per client, the stacking feature dimension is $\dim(z) = 2K = 6$, means each client contributes 2 features (one from CatBoost and one from Transformer).

This transformation represents a shift from the original feature space $\mathbb{R}^d$ to a model-output space $\mathbb{R}^{2K}$.

$$z(x) = [p_1^{\text{CB}}(x), p_1^{\text{TF}}(x), \dots, p_K^{\text{CB}}(x), p_K^{\text{TF}}(x)]^T \quad (27)$$

The final decision layer of the DP-FedHybrid framework, where a meta-learner, specifically LG, combines the outputs of all client-side models into a single prediction. In this step we transform the previously constructed meta-feature vector $\sigma(w^T z)$, into the final probability estimate described in Eq. (28), where $w \in \mathbb{R}^{2K}$ and $b \in \mathbb{R}$. This means that

the model computes a weighted sum of all meta-features, followed by a non-linear squashing function to produce a probabilistic output.

$$g(z) = \sigma(w^\top z + b) \quad (28)$$

The final prediction for an input x is calculated as in Eq. (29), which indicates that the prediction depends entirely on the meta-feature representation derived from all client models.

$$\hat{y}(x) = g(z(x)) \quad (29)$$

This demonstrates the final stage of the DP-FedHybrid model as a stacked LG classifier, where the input is not raw data but a vector of model predictions. The learned weights determine how much influence each client's CatBoost and Transformer outputs have on the final prediction. It makes the system a flexible and interpretable ensemble that integrates federated and hybrid learning principles.

In this work, we address energy efficiency by designing the framework with lightweight model components and limited computational overhead. The Transformer used in this work is intentionally configured with small embedding dimensions and shallow depth, significantly reducing computation compared to large-scale models. Also, the use of a few communication rounds (5 rounds) and limited local epochs helps minimize overall energy consumption during training. The inclusion of CatBoost, which is computationally efficient for tabular data, further reduces processing requirements. These designs collectively ensure that the proposed DP-FedHybrid framework remains energy-efficient and suitable for deployment on resource-constrained edge devices.

IV. EXPERIMENTS

The experimental evaluation of the proposed DP-FedHybrid framework is performed to assess the effectiveness of both standard FL and DP-SGD settings for heart disease prediction. The proposed pipeline is implemented using PyTorch for Transformer-based neural network modeling and CatBoost for gradient boosting-based learning on tabular data. Scikit-learn is utilized for data preprocessing, stacking-based meta-learning, and comprehensive performance evaluation. The experiments are conducted in a controlled environment to simulate a realistic decentralized healthcare scenario, where data is distributed across multiple clients without direct sharing. The dataset is first preprocessed by loading structured tabular features and corresponding binary labels representing the presence of heart disease. The data is split into 80% for training and 20% for testing, followed by a further partitioning of the training set into meta-training and meta-validation subsets. During the experiment, we observed that the meta-split is critical for enabling stacking-based meta-learning without introducing information leakage. To simulate a federated setting, the training data is divided into multiple client partitions using stratified sampling, ensuring that each client maintains a balanced class distribution. This setup reflects real-world

healthcare environments where data is distributed across institutions. Table 2 summarizes the key variables used in the proposed framework. These variables define the local and global model updates, gradient perturbation mechanisms, and privacy guarantees, providing a formal representation of the privacy-preserving FL process.

During model training, a global Transformer model is initialized with randomly assigned parameters. For each communication round, the global model parameters are broadcast to all participating clients. Each client then performs local training independently using its private dataset. The local training process consists of two parallel components. First, a CatBoost classifier is trained on the client's data using a fixed number of boosting iterations, depth, and learning rate. This model operates entirely locally and is not shared across clients. Second, a Transformer-based neural model is trained using mini-batch stochastic optimization. The Transformer employs a linear embedding layer followed by multiple residual feed-forward layers with non-linear activation and dropout, enabling it to learn complex feature interactions. During Transformer training, DP is enforced through a modified optimization procedure based on DP-SGD. For each mini-batch, gradients are computed with respect to the loss function, after which the total gradient norm is evaluated. To bound the sensitivity of the gradients, a clipping operation is applied such that the norm does not exceed a predefined threshold. Gaussian noise proportional to the clipping bound is added to the gradients before updating the model parameters. This process ensures that each client's contribution to the global model remains privacy-preserving. The local optimization is performed for a fixed number of epochs using the Adam optimizer, after which the updated model parameters are retained for aggregation. Once all clients complete their local updates, the server performs federated aggregation using the FedAvg algorithm. The parameters of the Transformer models from all clients are averaged element-wise to produce an updated global model. This aggregation step enables the model to capture shared patterns across decentralized datasets without accessing raw data. The updated global model is then redistributed to clients in the next round, and the process repeats for a predefined number of communication rounds. The proposed framework incorporates a stacking-based meta-learning stage to enhance predictive performance. After each round, predictions are generated for both CatBoost and Transformer models from all clients on the meta-training dataset. These predictions are treated as meta-features and concatenated to form a new feature space. The LG model is then trained on this meta-feature representation to learn an optimal combination of client-level predictions. This approach allows the model to leverage complementary strengths of different learners and improves robustness under heterogeneous data distributions. The same transformation is applied to validation and test data to obtain final predictions. Model evaluation is conducted at the end of each communication round using both the meta-validation set and the independent test set. Predictions

TABLE 2. Summary of experimental components and implementation details used in this study.

Component	Description
$\mathcal{D}$, Dataset	Complete tabular dataset with features and labels for heart disease prediction
x_i , Feature vector	Clinical input attributes of the i -th patient
y_i , Target label	Binary output indicating disease presence or absence
d , Feature size	Total number of input variables
N , Sample size	Total number of observations
K , Clients	Number of federated participants
$\mathcal{D}_k$, Local data	Dataset stored at client k
n_k , Local size	Number of samples at client k
f_k^{CB} , Tree model	CatBoost classifier for tabular learning
f_k^{TF} , Neural model	Transformer-based model
θ_k , Parameters	Local Transformer weights
$\theta^{(r)}$, Global model	Aggregated parameters at round r
R , Rounds	Number of FL communication iterations
E , Epochs	Local training iterations per round
B , Batch size	Samples per mini-batch
η , Learning rate	Step size for optimization
M , Iterations	Number of boosting steps in CatBoost
$h_m(x)$, Decision tree	Individual tree output in ensemble
p , Embedding dim	Transformer latent dimension
L , Layers	Number of Transformer blocks
$z^{(\ell)}$, Hidden state	Intermediate feature representation
$a_k(x)$, Logit	Pre-activation output of Transformer
$\sigma(\cdot)$, Activation	Sigmoid probability mapping
$\mathcal{L}_k$, Loss	Binary cross-entropy objective
g_k , Gradient	Parameter update direction
$\ g_k\ _2$, Norm	Gradient magnitude (L2)
C , Clip bound	Gradient clipping threshold
σ_{dp} , Noise scale	DP Gaussian noise multiplier
$\tilde{g}_k$, Noisy gradient	Perturbed gradient for privacy
$\theta_k^{(t+1)}$, Updated params	Parameters after DP-SGD update
β_1, β_2 , Adam factors	Exponential decay rates for moment estimation
m_t , First moment	Running average of gradients in Adam optimizer
v_t , Second moment	Running average of squared gradients
$\hat{m}_t, \hat{v}_t$, Bias-corrected	Adjusted moment estimates
τ , Stability term	Small constant to avoid division by zero
ϵ , Privacy budget	Measure of privacy guarantee strength
δ , Failure prob	Probability of privacy guarantee violation
α_k , Clip factor	Gradient scaling coefficient
w, b , Meta params	Logistic regression weights
$z(x)$, Meta features	Stacked prediction vector
$\hat{y}$, Prediction	Final output probability

are converted into binary labels using a fixed decision threshold, and standard classification metrics, including accuracy, precision, recall, and F1-score. To evaluate the impact of DP, experiments are performed under two settings: without privacy and with privacy enabled. The comparison highlights the trade-off between predictive performance and privacy preservation. The experimental results demonstrate that the proposed DP-FedHybrid framework achieves strong predictive performance while maintaining privacy guarantees. The federated hybrid design effectively captures both local and global data patterns, and the stacking mechanism enhances generalization across clients.

V. MODEL RESULTS

We conducted a series of experiments to comprehensively evaluate the performance of the proposed model under different settings. The model was evaluated using various metrics to ensure robustness, reliability, and generalization capability across diverse data distributions. Table 3 highlights the performance of centralized, FL, and DP-FL

configurations of the proposed model. In the centralized setting, the model achieves superior performance with 99.51% accuracy, 99.06% precision, 100% recall, and 99.53% F1-score, benefiting from full data accessibility and globally optimized feature representations. Using the federated setting with three clients, we obtained an accuracy of 95.12%, precision of 95.24%, a recall of 95.24%, and an F1-score of 95.24%. Using the DP-FL setting, up to five round, we obtained the results for the 5h round as an accuracy of 94.15%, precision of 92.66%, recall of 96.19%, and F1-score of 94.39%. The results demonstrate that the proposed framework maintains strong generalization and stability under decentralized constraints using standard FL and DP-FL settings. Table 4, indicates model performance across early rounds, reflecting stable local optimization and convergence. Performance improvement in later rounds demonstrates enhanced global aggregation via FedAvg, leading to better recall and F1-score. This suggests effective cross-client knowledge integration and improved generalization under iterative noise variance.

TABLE 3. Performance of the proposed model using both centralized and FL settings.

Model	Setting	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
CatBoost + Transformer	Centralized	99.51	99.06	100	99.53
CatBoost + Transformer	Federated (3 Clients)	95.12	95.24	95.24	95.24
CatBoost + Transformer	DP-FL (3 Clients)	94.15	92.66	96.19	94.39

The performance of the proposed DP-FedHybrid is attributed to its multi-level learning strategy and robust aggregation approach. At the local level, CatBoost effectively captures high-order non-linear feature interactions, while the Transformer learns latent feature representations and contextual dependencies, enabling complementary modeling of tabular data. The federated optimization of the Transformer via FedAvg facilitates cross-client knowledge sharing, mitigating overfitting to local distributions and enhancing global generalization. Also, the incorporation of DP-SGD introduces controlled stochastic perturbations, which act as a regularization mechanism, preventing sharp minima convergence while preserving privacy. At the aggregation level, the stacking-based meta-learner learns optimal weighting of heterogeneous client predictions, effectively reducing client drift and variance induced by non-IID data. This hierarchical fusion of predictions improves decision boundary calibration compared to single-model FL approaches.

A. ABLATION STUDY

In this work, we performed various ablation studies to evaluate the impact of client participation and communication rounds by analyzing single-client, pairwise, and full-client federated configurations. The obtained results demonstrate that multi-client collaboration significantly improves generalization through cross-client knowledge aggregation. Also, the DP ablations across varying noise levels reveal robust performance stability under low perturbation, with controlled degradation at higher noise, validating the model's privacy-utility trade-off efficiency. The DP-FL results illustrated in Table 5 show stable performance across initial rounds, indicating effective privacy-preserving optimization with controlled gradient perturbation. The slight fluctuations and the final decline in accuracy reflect the impact of cumulative noise injection. Also, the consistently high recall demonstrates robust sensitivity, demonstrating that DP-SGD maintains predictive reliability under privacy constraints.

Table 6 illustrates the model sensitivity to varying DP noise levels under fixed clipping. Performance remains invariant up to noise = 0.3, indicating robustness to moderate perturbations. A slight degradation at noise = 0.2 suggests threshold-dependent instability. These results demonstrate that DP-SGD preserves utility under controlled noise while maintaining stable precision-recall characteristics.

Table 7 evaluates the impact of client participation and communication rounds using the proposed model. During the experiments, we observed that the single-client setups show lower generalization, while multi-client configurations improve performance via collaborative learning. The

full-client setting achieves stable optimal metrics, indicating effective knowledge aggregation.

Figure 3 illustrates the privacy-utility behavior of DP-FedHybrid. The proposed model shows a stable accuracy of 95.12% and an F1-score of 95.24%, under low-to-moderate noise at 0.2. The performance dip at intermediate noise 0.4 indicates sensitivity to gradient perturbation, followed by recovery, suggesting robustness and a non-linear response to DP noise scaling.

Figure 4 illustrates the class-wise predictive probability distributions for Standard FL and DP-FL. Both models demonstrate strong class separability, with negative samples concentrated near 0 and positive samples near 1. The proposed DP-FL preserves this separation with minimal dispersion, indicating that DP does not degrade discriminative capability for heart disease prediction.

The proposed study describes an FL strategy that takes into consideration data heterogeneity by allowing for diverse model designs among clients. It also allows for unified prediction by employing a stacking-based meta-optimization method. The model uses complementary inductive biases by integrating gradient boosting for non-linear partitioning and Transformer-based representation learning under decentralized constraints. DP-SGD enforces bounded gradient sensitivity via norm clipping. It introduces calibrated stochastic perturbations, enabling privacy-preserving optimization while maintaining stable convergence dynamics. The framework further advances FL by demonstrating robust convergence under non-IID data distributions and efficient communication with minimal rounds.

B. RESULTS COMPARISON

Existing studies on heart disease prediction commonly used either classic ML models, such as (SVM) or DL models such as Artificial Neural Networks (ANN), often in centralized settings. Recent studies have used FL for data privacy but largely they used homogeneous model architectures, limiting representational diversity. Table 8 illustrates the closely related works for heart disease prediction using both centralized and FL approaches. The study conducted in [1] used a DL approach for heart disease prediction. They used a dense neural network by applying different architectures with 3 to 9 hidden layers, where each layer consisted of 100 neurons with ReLU activation. They achieved an accuracy of 83%. Their model works well; however, a key limitation of their model is that it depends on fixed network configurations without incorporating feature selection or optimization techniques, which may restrict its overall predictive performance. The study conducted

TABLE 4. Performance metrics of the proposed DP-FedHybrid model across five FL rounds.

Round	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Round 1	95.12	94.39	96.19	95.28
Round 2	95.12	94.39	96.19	95.28
Round 3	95.12	94.39	96.19	95.28
Round 4	95.61	94.44	97.14	95.77
Round 5	95.61	94.44	97.14	95.77

TABLE 5. Performance metrics of the proposed DP-FedHybrid model using DP-FL setting.

Round	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Round 1	95.12	94.39	96.19	95.28
Round 2	95.61	94.44	97.14	95.77
Round 3	95.12	94.39	96.19	95.28
Round 4	95.12	94.39	96.19	95.28
Round 5	94.15	92.66	96.19	94.39

TABLE 6. Impact of DP noise levels on the proposed DP-FedHybrid model performance.

Setting	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
DP-FL noise=0.0, clip=1.0	95.12	95.24	95.24	95.24
DP-FL noise=0.02, clip=1.0	95.12	95.24	95.24	95.24
DP-FL noise=0.05, clip=1.0	95.12	95.24	95.24	95.24
DP-FL noise=0.3, clip=1.0	95.12	95.24	95.24	95.24
DP-FL noise=0.2, clip=1.0	94.63	94.34	95.24	94.79

TABLE 7. Client subset and communication round ablation analysis in federated hybrid learning.

Client Subset	Num Clients	Rounds	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
(1)	1	1	89.27	87.39	92.38	89.81
(1)	1	3	89.27	87.39	92.38	89.81
(1)	1	5	89.27	87.39	92.38	89.81
(2)	1	1	87.32	86.24	89.52	87.55
(2)	1	3	86.34	85.98	87.62	86.79
(2)	1	5	86.34	85.98	87.62	86.79
(3)	1	1	94.63	93.52	96.19	94.84
(3)	1	3	94.15	93.46	95.24	94.34
(3)	1	5	94.63	93.52	96.19	94.84
(1,2)	2	1	94.63	94.34	95.24	94.79
(1,2)	2	3	94.63	94.34	95.24	94.79
(1,2)	2	5	94.63	94.34	95.24	94.79
(1,3)	2	1	93.66	92.59	95.24	93.90
(1,3)	2	3	95.12	92.04	99.05	95.41
(1,3)	2	5	94.63	92.73	97.14	94.88
(2,3)	2	1	92.68	94.12	91.43	92.75
(2,3)	2	3	92.68	94.12	91.43	92.75
(2,3)	2	5	93.17	95.05	91.43	93.20
(1, 2, 3)	3	1	95.12	95.24	95.24	95.24
(1, 2, 3)	3	3	95.12	95.24	95.24	95.24
(1, 2, 3)	3	5	95.12	95.24	95.24	95.24

in [37] used a pre-trained Deep Neural Network (DNN) for feature extraction, Principal Component Analysis (PCA) for dimensionality reduction, and LR for classification. They evaluated their model on the publicly available heart disease dataset and achieved an accuracy of 93.33%. Their method works well; however, a key limitation is its reliance on pre-trained features, which may limit adaptability to dataset-specific patterns and variations. The study conducted in [36] used a Grey Wolf Horse Herd Optimization-based Shepard Convolutional Neural Network (GWHHO-ShCNN), for heart disease prediction. They integrate Spark-based architecture, feature fusion using Hellinger distance and Deep Q Network

(DQN). Using the heart disease dataset, they achieved an accuracy of 93.25%. They achieved reasonable accuracy; however, their model's limitation is its high computational complexity due to multiple optimization and distributed processing.

For heart disease prediction, the study in [38] used various methods such as enhanced entropy-based features, statistical features, and information gain features, which are then used as input to an Improved Quantum Convolutional Neural Network (IQCNN) for classification. They achieved an accuracy of 92.3%, and an F1-score of 90.14%. However, a key limitation is the high computational complexity of the

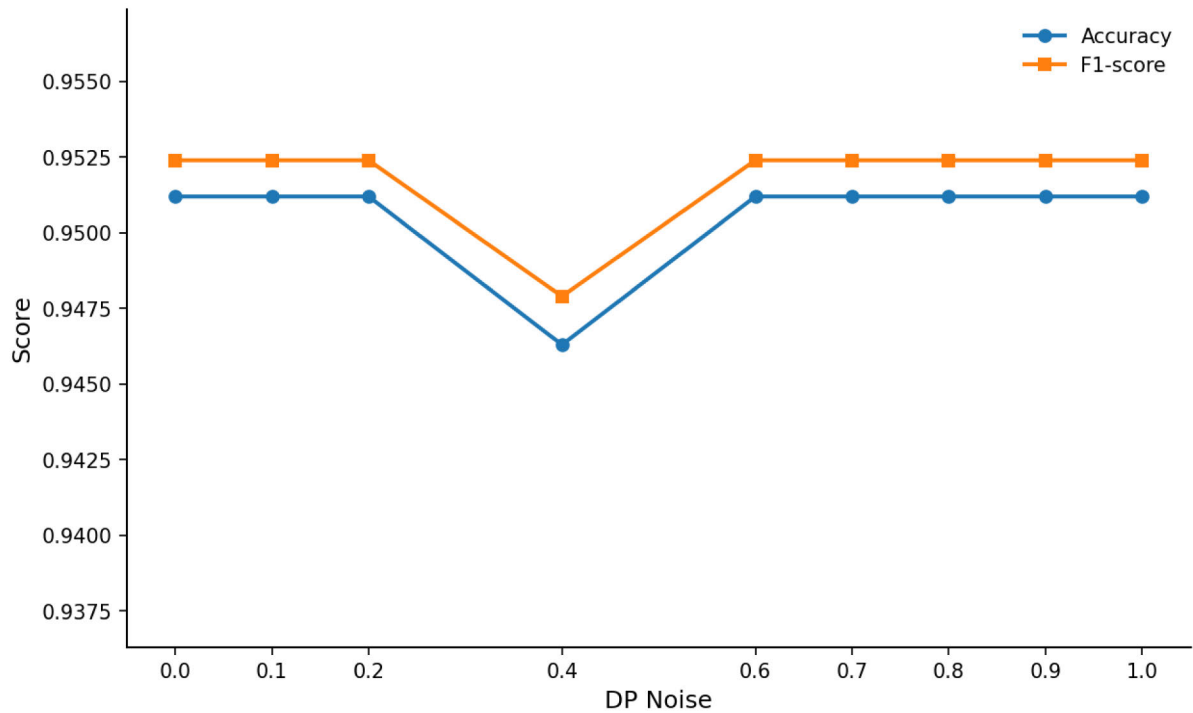


FIGURE 3. Privacy-utility tradeoff of the proposed DP-FedHybrid model under varying DP noise levels.

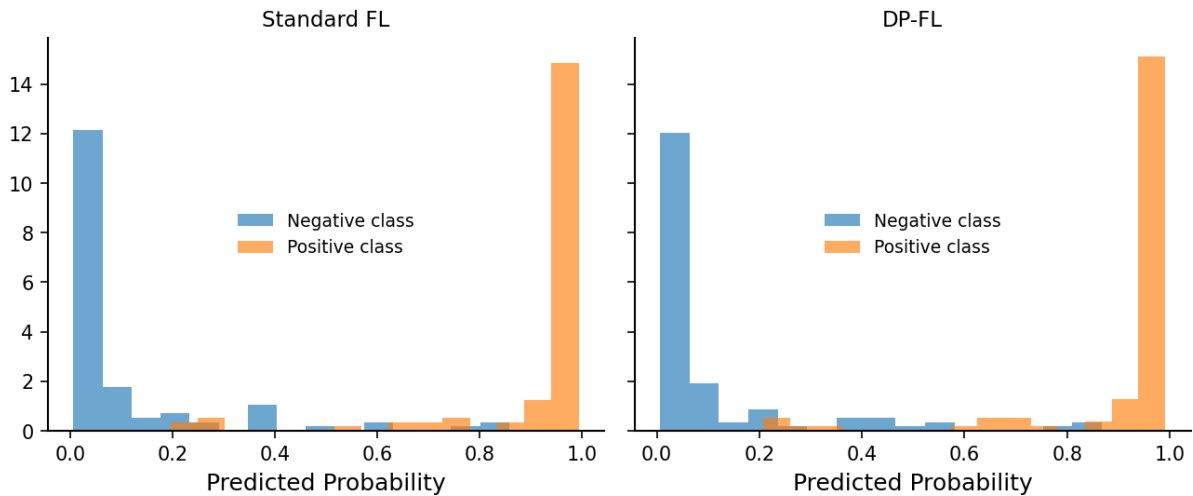


FIGURE 4. Class-wise predictive probability distributions using both standard FL and DP-FL.

IQCNN model, which may restrict its practical deployment in real-world clinical settings. The study conducted in [39] combines a Stacking Classifier (SC) with an improved Binary Salp Swarm Algorithm (BSSA) enhanced by wolf optimization and opposition-based learning for feature selection. They obtained an accuracy of 95% and an F1-score of 95%. Their model works well; however, a key limitation is the increased model complexity and computational overhead due to the integration of multiple optimizations and ensemble learning components. The study in [40] used an FL framework,

FedHFP, for heart failure prediction. They used various models such as ANN, CNN, Recurrent Neural Network (RNN), Long Short-Term Memory (LSTM) network, and Gated Recurrent Unit (GRU) under different network settings. They performed various experiments, where the ANN model achieved the best performance with an accuracy of 93.75%.

The study in [3] used a privacy-preserving disease prediction framework by integrating FL with feature selection (ANOVA and Chi-Square) and feature extraction using

TABLE 8. Results comparison of the proposed DP-FedHybrid model with closely related works.

Study	Dataset	Model	F1-score (%)	Accuracy (%)
Almazroi et al. [1], 2023	Heart Disease	Dense neural network	–	83
Kanchanamala et al. [36], 2023	Heart Disease	GWHHO-based ShCNN	–	93.25
Hassan et al. [37], 2023	Heart Disease	DNN + PCA + LR	–	93.33
Pitchal et al. [38], 2024	Heart Disease	IQCNN	90.14	92.3
Sowmiya et al. [39], 2025	Heart Disease	BSSA + SC	95	95
Lohachab et al. [40], 2025	Heart Disease	FedHFP, ANN	–	93.75
Kapila et al. [3], 2025	Heart Disease	FL, Anova and Chi-Square	89.23	88.52
Proposed work, 2026	Heart Disease	DP-FedHybrid, CatBoost + Transformer + Stacking	95.24	95.12

Linear Discriminant Analysis (LDA). Using the heart disease dataset, they obtained an accuracy of 88.52% and an F1-score of 89.23%. Their method performed well; however, the model is limited by potential communication overhead and increased training complexity associated with FL across multiple distributed institutions.

Our proposed DP-FedHybrid framework achieved better performance through multi-level representation fusion and decentralized ensemble optimization. During the experiments, we observed that integrating CatBoost's capability to capture high-order non-linear feature interactions with the Transformer's latent representation learning enhances feature expressiveness under tabular constraints. The stacking-based meta-learner further optimizes prediction by learning adaptive weighting of heterogeneous client outputs, reducing bias and variance. Federated aggregation via FedAvg enables cross-client knowledge generalization, while DP-SGD introduces controlled stochastic regularization. Our proposed model obtained an accuracy of 95.12% and an F1-score of 95.24% under FL, and maintains 94.15% accuracy under DP, demonstrating an effective privacy-utility trade-off. Also, the use of FedAvg aggregation, lightweight Transformer architecture, and efficient CatBoost models allows the framework to extend to a larger number of clients without significant computational overhead.

VI. DISCUSSIONS

Our proposed DP-FedHybrid framework demonstrates a significant advancement using an FL setup for heart disease prediction by effectively addressing key limitations of prior approaches. Unlike conventional FL methods that depend on homogeneous architectures, the integration of tree-based and transformer models enables the capture of both non-linear feature interactions and high-dimensional contextual dependencies, resulting in enhanced representational capacity. The stacking-based meta-learning layer further mitigates the impact of client-level heterogeneity by adaptively combining probabilistic outputs, thereby improving robustness and generalization under non-IID data distributions. The key contribution of the model is the incorporation of DP within the transformer optimization process. By applying gradient clipping and calibrated noise injection at the client level, the proposed framework ensures formal privacy guarantees while maintaining stable convergence behavior. Unlike traditional DP-FL implementations, the model demonstrates resilience

to moderate noise injection, suggesting that hybridization inherently regularizes the learning process and reduces sensitivity to perturbations.

In this work, we addressed communication efficiency by designing the framework to operate with lightweight model updates and limited communication rounds. Specifically, the Transformer component is intentionally kept shallow with a small embedding dimension, reducing the size of transmitted parameters. Also, our proposed system uses a small number of communication rounds (5 rounds), which minimizes communication frequency while still achieving convergence. The use of stacking-based meta-learning further reduces reliance on continuous global updates by leveraging local model predictions. These design choices collectively help in lowering communication overhead, making the proposed framework more practical for real-world federated deployments. The stacking-based meta-learning layer enhances the framework by introducing adaptive cross-client knowledge fusion, which reduces client drift and improves robustness under non-IID distributions. Ablation analyses confirm that collaborative configurations outperform isolated clients, while convergence studies indicate that the model achieves early stabilization with minimal communication overhead, highlighting its efficiency. The proposed framework establishes a scalable and privacy-preserving hybrid learning paradigm, demonstrating that combining heterogeneous models with meta-level aggregation can significantly enhance performance in decentralized, sensitive data environments.

VII. CONCLUSION

In this work, we present DP-FedHybrid, a novel FL framework that systematically integrates heterogeneous model paradigms under strict privacy constraints for heart disease prediction. The proposed approach is unique from conventional homogeneous FL designs by introducing a prediction-level hybridization strategy, where tree-based and transformer-based models are jointly leveraged through a stacking-based meta-learning mechanism. This design enables the model to exploit complementary inductive biases, resulting in improved representational learning and robustness in non-IID environments. The key contribution of this work lies in the selective incorporation of DP within the transformer optimization process. It enables the framework to preserve sensitive clinical information

while maintaining stable convergence. Unlike traditional DP-FL approaches, the proposed framework achieves a better privacy-utility trade-off, demonstrating minimal degradation in predictive performance. Also, the integration of a meta-level aggregation layer enhances cross-client knowledge fusion beyond standard parameter averaging, effectively mitigating client drift and improving generalization. The empirical results confirm that the proposed framework consistently outperforms standalone and conventional federated models, particularly in heterogeneous and privacy-sensitive environments. This study contributes a scalable, privacy-aware, and hybrid learning paradigm that advances the state-of-the-art in FL and provides a practical foundation for secure deployment of predictive models in real-world clinical systems.

Future work will focus on extending the proposed framework to larger-scale multi-institutional datasets and incorporating adaptive privacy mechanisms to dynamically balance privacy and utility. Also, integrating communication-efficient federated optimization and advanced meta-learning strategies can further enhance scalability and generalization.

REFERENCES

- [1] A. A. Almazroi, E. A. Aldahri, S. Bashir, and S. Ashfaq, "A clinical decision support system for heart disease prediction using deep learning," *IEEE Access*, vol. 11, pp. 61646–61659, 2023.
- [2] M. S. A. Reshan, S. Amin, M. A. Zeb, A. Sulaiman, H. Alshahrani, and A. Shaikh, "A robust heart disease prediction system using hybrid deep neural networks," *IEEE Access*, vol. 11, pp. 121574–121591, 2023.
- [3] R. Kapila and S. Saleti, "Federated learning-based disease prediction: A fusion approach with feature selection and extraction," *Biomed. Signal Process. Control*, vol. 100, Feb. 2025, Art. no. 106961.
- [4] S. Alasmari, R. AlGhamdi, G. G. Tejani, S. Kumar Sharma, and S. J. Mousavirad, "Federated learning-based multimodal approach for early detection and personalized care in cardiac disease," *Frontiers Physiol.*, vol. 16, Apr. 2025, Art. no. 1563185.
- [5] X. Wang, J. Hu, H. Lin, W. Liu, H. Moon, and M. J. Piran, "Federated learning-empowered disease diagnosis mechanism in the Internet of Medical Things: From the privacy-preservation perspective," *IEEE Trans. Ind. Inform.*, vol. 19, no. 7, pp. 7905–7913, Jul. 2023.
- [6] A. U. Rahman, Y. Alsenani, A. Zafar, K. Ullah, K. Rabie, and T. Shongwe, "Enhancing heart disease prediction using a self-attention-based transformer model," *Sci. Rep.*, vol. 14, no. 1, p. 514, Jan. 2024.
- [7] S. Shajimon, R. M. Shukla, and A. N. Patra, "Precision at heart: An IoT-based vertical federated learning approach for heterogeneous data-driven cardiovascular disease risk prediction," *Comput. Methods Programs Biomed.*, vol. 273, Jan. 2026, Art. no. 109079.
- [8] T. Deng, H. Hamdan, R. Yaakob, and K. A. Kasmiran, "Personalized federated learning for in-hospital mortality prediction of multi-center ICU," *IEEE Access*, vol. 11, pp. 11652–11663, 2023.
- [9] M. Karami and A. Karami, "Harmony in federated learning: A comprehensive review of techniques to tackle heterogeneity and non-IID data," *Cluster Comput.*, vol. 28, no. 9, p. 570, Oct. 2025.
- [10] H. Yang, M. Ge, K. Xiang, X. Bai, and H. Li, "FedVAE: Communication-efficient federated learning with non-IID private data," *IEEE Syst. J.*, vol. 17, no. 3, pp. 4798–4808, Sep. 2023.
- [11] W. Qiu, Y. Feng, Y. Li, Y. Chang, K. Qian, B. Hu, Y. Yamamoto, and B. W. Schuller, "Fed-MStacking: Heterogeneous federated learning with stacking misaligned labels for abnormal heart sound detection," *IEEE J. Biomed. Health Informat.*, vol. 28, no. 9, pp. 5055–5066, Sep. 2024.
- [12] M. Ali, F. Naeem, M. Tariq, and G. Kaddoum, "Federated learning for privacy preservation in smart healthcare systems: A comprehensive survey," *IEEE J. Biomed. Health Informat.*, vol. 27, no. 2, pp. 778–789, Feb. 2023.
- [13] A. Linardos, K. Kushibar, S. Walsh, P. Gkontra, and K. Lekadir, "Federated learning for multi-center imaging diagnostics: A simulation study in cardiovascular disease," *Sci. Rep.*, vol. 12, no. 1, p. 3551, Mar. 2022.
- [14] M. Tölle et al., "Real world federated learning with a knowledge distilled transformer for cardiac CT imaging," *npj Digit. Med.*, vol. 8, no. 1, p. 88, Feb. 2025.
- [15] M. Wei, J. Yang, Z. Zhao, X. Zhang, J. Li, and Z. Deng, "DeFedHDP: Fully decentralized online federated learning for heart disease prediction in computational health systems," *IEEE Trans. Computat. Social Syst.*, vol. 11, no. 5, pp. 6854–6867, Oct. 2024.
- [16] S. Molaei, A. Thakur, L. Clifton, A. Soltan, P. Schwab, D. Belgrave, K. Branson, and D. A. Clifton, "Learning across the divide: Personalised federated learning for robust clinical modelling under data-view heterogeneity," *IEEE J. Biomed. Health Informat.*, vol. 30, no. 3, pp. 2000–2009, Mar. 2026.
- [17] B. Ramesh and K. Lakshmana, "A novel early detection and prevention of coronary heart disease framework using hybrid deep learning model and neural fuzzy inference system," *IEEE Access*, vol. 12, pp. 26683–26695, 2024.
- [18] J. Chen, S. Huang, Y. Zhang, Q. Chang, Y. Zhang, D. Li, J. Qiu, L. Hu, X. Peng, Y. Du, Y. Gao, D. Z. Chen, A. Bellou, J. Wu, and H. Liang, "Congenital heart disease detection by pediatric electrocardiogram based deep learning integrated with human concepts," *Nature Commun.*, vol. 15, no. 1, p. 976, Feb. 2024.
- [19] S. González, A. K.-C. Yi, W.-T. Hsieh, W.-C. Chen, C.-L. Wang, V. C.-C. Wu, and S.-H. Chang, "Multi-modal heart failure risk estimation based on short ECG and sampled long-term HRV," *Inf. Fusion*, vol. 107, Jul. 2024, Art. no. 102337.
- [20] H. El-Sofany, B. Bouallegue, and Y. M. A. El-Latif, "A proposed technique for predicting heart disease using machine learning algorithms and an explainable AI method," *Sci. Rep.*, vol. 14, no. 1, p. 23277, Oct. 2024.
- [21] A. M. Qadri, A. Raza, K. Munir, and M. S. Almutairi, "Effective feature engineering technique for heart disease prediction with machine learning," *IEEE Access*, vol. 11, pp. 56214–56224, 2023.
- [22] S. Santhosh, K. Chadaga, R. V. Arjunan, and S. D'Souza, "Cardiac clarity: Harnessing machine learning for accurate heart-disease prediction," *IEEE Access*, vol. 13, pp. 97529–97544, 2025.
- [23] K. K. Baseer, K. Sivakumar, D. Veeraiah, G. Chhabra, P. Kumar Lakineni, M. Jahir Pasha, R. Gandikota, and G. Harikrishnan, "Healthcare diagnostics with an adaptive deep learning model integrated with the Internet of Medical Things (IoMT) for predicting heart disease," *Biomed. Signal Process. Control*, vol. 92, Jun. 2024, Art. no. 105988.
- [24] A. Ashtaiwi, T. Khalifa, and O. Alirri, "Enhancing heart disease diagnosis through ECG image vectorization-based classification," *Heliyon*, vol. 10, no. 18, Sep. 2024, Art. no. e37574.
- [25] W. Qiu, C. Quan, L. Zhu, Y. Yu, Z. Wang, Y. Ma, M. Sun, Y. Chang, K. Qian, B. Hu, Y. Yamamoto, and B. W. Schuller, "Heart sound abnormality detection from multi-institutional collaboration: Introducing a federated learning framework," *IEEE Trans. Biomed. Eng.*, vol. 71, no. 10, pp. 2802–2813, Oct. 2024.
- [26] G. Shen, Z. Fu, Y. Gui, W. Susilo, and M. Zhang, "Efficient and privacy-preserving online diagnosis scheme based on federated learning in e-healthcare system," *Inf. Sci.*, vol. 647, Nov. 2023, Art. no. 119261.
- [27] F. S. Alreshidi, M. Alsaffar, R. Chengoden, and N. K. Alshammari, "Fed-CL- an atrial fibrillation prediction system using ECG signals employing federated learning mechanism," *Sci. Rep.*, vol. 14, no. 1, p. 21038, Sep. 2024.
- [28] N. Naveenkumar, T. G. Subarna, S. Pragadeeswaran, and V. Karupuchamy, "Federated learning based hybrid dlbased transformer model for heart disease detection with explainable AI," *Biomed. Signal Process. Control*, vol. 120, Jul. 2026, Art. no. 110076.
- [29] R. N. Asif, A. Ditta, H. Alquhayz, S. Abbas, M. A. Khan, T. M. Ghazal, and S.-W. Lee, "Detecting electrocardiogram arrhythmia empowered with weighted federated learning," *IEEE Access*, vol. 12, pp. 1909–1926, 2024.
- [30] M. Dubey, J. Tembhurne, and R. Makhijani, "Enhancing federated learning through differential privacy: Introducing FedHybrid for multicenter diverse heart disease datasets," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 9, no. 6, pp. 4217–4230, Dec. 2025.
- [31] S. Alammari, H. Aldawsari, A. AlSahly, K. A. AlGhamdi, and M. A. Khan, "Federated explainable AI for privacy-preserving cardiovascular disease detection using wearable devices," *IEEE Trans. Consum. Electron.*, vol. 72, no. 1, pp. 2164–2175, Feb. 2026.

- [32] M. A. Noor, S. B. Hassan, M. S. Bin Alam, A. Lameesa, and M. A. R. Siddique, "A privacy-preserving federated learning framework with graph neural networks for enhanced heart attack risk prediction," *Array*, vol. 27, Sep. 2025, Art. no. 100500.
- [33] S. Rani, A. Kataria, S. Kumar, and P. Tiwari, "Federated learning for secure IoMT-applications in smart healthcare systems: A comprehensive review," *Knowledge-Based Syst.*, vol. 274, Aug. 2023, Art. no. 110658.
- [34] F. Ali, S. El-Sappagh, S. M. R. Islam, D. Kwak, A. Ali, M. Imran, and K.-S. Kwak, "A smart healthcare monitoring system for heart disease prediction based on ensemble deep learning and feature fusion," *Inf. Fusion*, vol. 63, pp. 208–222, Nov. 2020.
- [35] A. Janosi, W. Steinbrunn, M. Pfisterer, and R. Detrano, "Heart disease [dataset]," UC Irvine Mach. Learn. Repository, Cleveland, OH, USA, Tech. Rep., 1989, doi: [10.24432/C52P4X](https://archive.ics.uci.edu/dataset/45/heart+disease). [Online]. Available: <https://archive.ics.uci.edu/dataset/45/heart+disease>
- [36] P. Kanchanamala, A. S. Alphonse, and P. V. B. Reddy, "Heart disease prediction using hybrid optimization enabled deep learning network with spark architecture," *Biomed. Signal Process. Control*, vol. 84, Jul. 2023, Art. no. 104707.
- [37] D. Hassan, H. I. Hussein, and M. M. Hassan, "Heart disease prediction based on pre-trained deep neural networks combined with principal component analysis," *Biomed. Signal Process. Control*, vol. 79, Jan. 2023, Art. no. 104019.
- [38] P. Pitchal, S. Ponnusamy, and V. Soundararajan, "Heart disease prediction: Improved quantum convolutional neural network and enhanced features," *Expert Syst. Appl.*, vol. 249, Sep. 2024, Art. no. 123534.
- [39] M. Sowmiya, B. Banu Rekha, and E. Malar, "Optimized heart disease prediction model using a meta-heuristic feature selection with improved binary salp swarm algorithm and stacking classifier," *Comput. Biol. Med.*, vol. 191, Jun. 2025, Art. no. 110171.
- [40] A. Lohachab and K. Kumar, "FedHFP: A federated deep learning framework for heart failure prediction," *IETE J. Res.*, vol. 71, no. 2, pp. 479–491, Feb. 2025.



ATTA UR RAHMAN received the Ph.D. degree in computer science, with a specialization in artificial intelligence from the Ghulam Ishaq Khan Institute of Engineering Sciences and Technology (GIKI), Pakistan, in 2022. He is currently a Postdoctoral Research Fellow with the King Fahd University of Petroleum and Minerals (KFUPM), Saudi Arabia. He is working on the design and deployment of advanced AI-driven frameworks for healthcare, breast cancer detection, brain–computer interfaces (BCI), and human–computer interaction (HCI), with a strong emphasis on transformer architectures, federated learning, and multimodal deep learning. He has authored more than 40 publications in various well-reputed journals, such as *Biomedical Signal Processing and Control*, *IEEE TRANSACTIONS*, *IEEE ACCESS*, and *Scientific Reports*. His research integrates computer vision, NLP, graph neural networks, and signal processing (EEG and ECG) to develop clinically robust diagnostic systems. He works on attention mechanisms, representation learning, and non-IID data challenges to enhance model generalization, interpretability, and real-world scalability.



BIBI SAQIA received the M.S. degree in computer science and the Ph.D. degree in computer science, with a specialization in artificial intelligence from the University of Science and Technology Bannu, Bannu, Pakistan, in 2018. She has authored various publications in well-reputed journals and conferences, including *Biomedical Signal Processing and Control* and *IEEE ACCESS*. She serves as a reviewer for several international journals. Her research interests include data mining and artificial intelligence in healthcare, NLP, and computational intelligence.



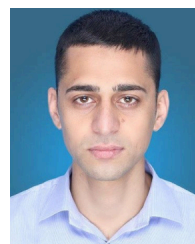
MUHAMMAD TAYYAB received the Ph.D. degree in industrial and management engineering from Hanyang University, South Korea. He is currently an Assistant Professor of operations management with the Information Systems and Operations Management Department, King Fahd Business School, King Fahd University of Petroleum and Minerals (KFUPM), Saudi Arabia. His research interests include fuzzy inference systems, machine learning, NLP, multi-stage production management, and supply chain risk management.



MEHRAN ULLAH received the Ph.D. degree in industrial management engineering. He is currently a Lecturer with the School of Business and Creative Industries, University of the West of Scotland, U.K. He is working on various projects, such as economic assessment of imperfect serial production management under uncertainty and stacked encoded cascade error feedback deep extreme learning networks for manufacturing order completion time. His research interests include operations research, supply chain management, inventory management, logistics, and sustainable supply chains.



MUHAMMAD SHAHID ANWAR (Senior Member, IEEE) received the M.Sc. degree in telecommunications technology from Aston University, Birmingham, U.K., and the Ph.D. degree in information and communication engineering from Beijing Institute of Technology, China, in 2021. He is currently with the King Fahd University of Petroleum and Minerals (KFUPM), Saudi Arabia. His research interests include artificial intelligence (AI), machine learning (ML), virtual reality (VR), augmented reality (AR), extended reality (XR), and the metaverse.



PIR NOMAN AHMAD received the B.S. degree (Hons.) in computer science from the University of Malakand, Pakistan, and the M.S. and Ph.D. degrees in computer science from Harbin Institute of Technology, China. He is currently with the Interdisciplinary Research Center for Finance and Digital Economy, King Fahd University of Petroleum and Minerals (KFUPM), Saudi Arabia, where he is a Researcher and the Co-Principal Investigator on interdisciplinary projects. His research interests include artificial intelligence, biomedical data analysis, and information systems, with a particular focus on AI-driven healthcare applications. He is a member of ACM.

...